Entropy-stabilized or kinetically trapped? Why some multi-metal hydroxides co-precipitate as one phase

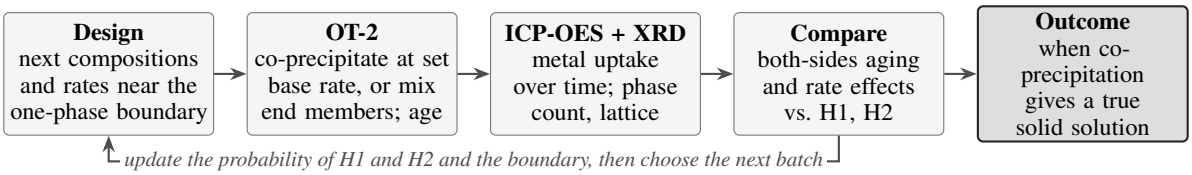
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Research question. Hydroxides and layered double hydroxides with four to six metals ("high-entropy" hydroxides) are made by room-temperature co-precipitation for oxygen-evolution catalysts, battery-cathode precursors and supercapacitors, and many are reported as a single phase. Why do some cation sets form one phase while others segregate?

Knowledge gap. For oxides made at high temperature, entropy stabilization has a clear test: the single phase forms reversibly on heating and separates on cooling [1]. At room temperature, the configurational entropy of five equimolar cations is worth only $TR\ln 5 \approx 4$ kJ per mole of cations, about a quarter of its value where the first entropy-stabilized oxide forms, and small enough for a modest mixing enthalpy to outweigh. A room-temperature single phase may instead be trapped by fast, simultaneous precipitation. Libraries made on liquid handlers [2,3] map which compositions give one phase, but not why.

Hypotheses. *H1, equilibrium:* the single phase is the stable state. Predictions: a physical mixture of separately precipitated end members, aged in its mother liquor, converges by dissolution and reprecipitation to the same single phase, and the single-phase region does not depend on how fast base is added. *H2, kinetic trapping:* the single phase exists because all cations precipitated together. Predictions: co-precipitates unmix on aging, faster when warmer; mixtures stay mixed; and the single-phase region shrinks when base is added slowly, so that cations precipitate in order of solubility.

Approach. Cation sets are chosen so that mismatches in size, solubility and preferred structure vary independently. Ni^{2+} and Mg^{2+}, for example, have radii within 5% but hydroxide solubility products about 10^4 apart; Cu^{2+} and Zn^{2+}, whose hydroxides are not brucite-type, add a structural mismatch. The sets build up to (Mg,Co,Ni,Cu,Zn), the cations of the first entropy-stabilized oxide [1]. An Opentrons OT-2 dispenses metal-salt solutions and base at set addition rates into vials, prepares the matching end-member mixtures, and ages both at room temperature and up to 95 °C on its heater-shaker, sampling the supernatant for ICP-OES to follow each metal's uptake and exchange over time. Powder XRD gives the number of phases and lattice parameters for Vegard's-law checks. Because XRD limits throughput, Bayesian active learning places samples near the single-phase boundary, where H1 and H2 predict different outcomes.



Expected outcomes. (1) Single-phase boundaries approached from both sides, as a function of composition, base-addition rate and aging temperature; (2) the rate at which each cation exchanges between solid and solution; (3) a causal rule for when co-precipitation gives a solid solution. That says whether a reported "high-entropy" hydroxide is stable, which governs how it transforms in an electrode or electrolyte.

Fit and resources. The project asks a "why" question about the *mechanism of chemical synthesis* under *non-equilibrium conditions*, with *data-driven* sample selection. It does not tune properties, optimize a synthesis or study catalytic mechanisms. The PI works on self-driving labs and Bayesian experimental design; the lab's OT-2 runs aqueous precipitation series; campus ICP-OES and XRD are available.

References. [1] C. M. Rost *et al.*, *Nat. Commun.* 6, 8485 (2015), doi:10.1038/ncomms9485. [2] L. Velasco *et al.*, *Adv. Mater.* 33, 2102301 (2021), doi:10.1002/adma.202102301. [3] Zeinali Galabi *et al.*, *Adv. Mater.* 38 (2026), doi:10.1002/adma.202519790.